

OSCE Prep

Hematology

Hematology

Anemia

Microcytic anemias

- Iron deficiency anemia
- Lead poisoning
- Thalassemias
- Sideroblastic anemia
- Anemia of chronic illness

Macrocytic anemia's

- Medications (eg, anti-convulsants, zidovudine, and immunosuppressive agents)
- Vitamin B12 or folate deficiency or inborn errors of metabolism
- liver disease
- Aplastic anemia

Normocytic anemia's

- Decreased production Aplastic anemias
- Hemorrhage (recent)
- Sequestration
- Infection
- Renal disease

The previous all
are associated
with decreased
RBC production

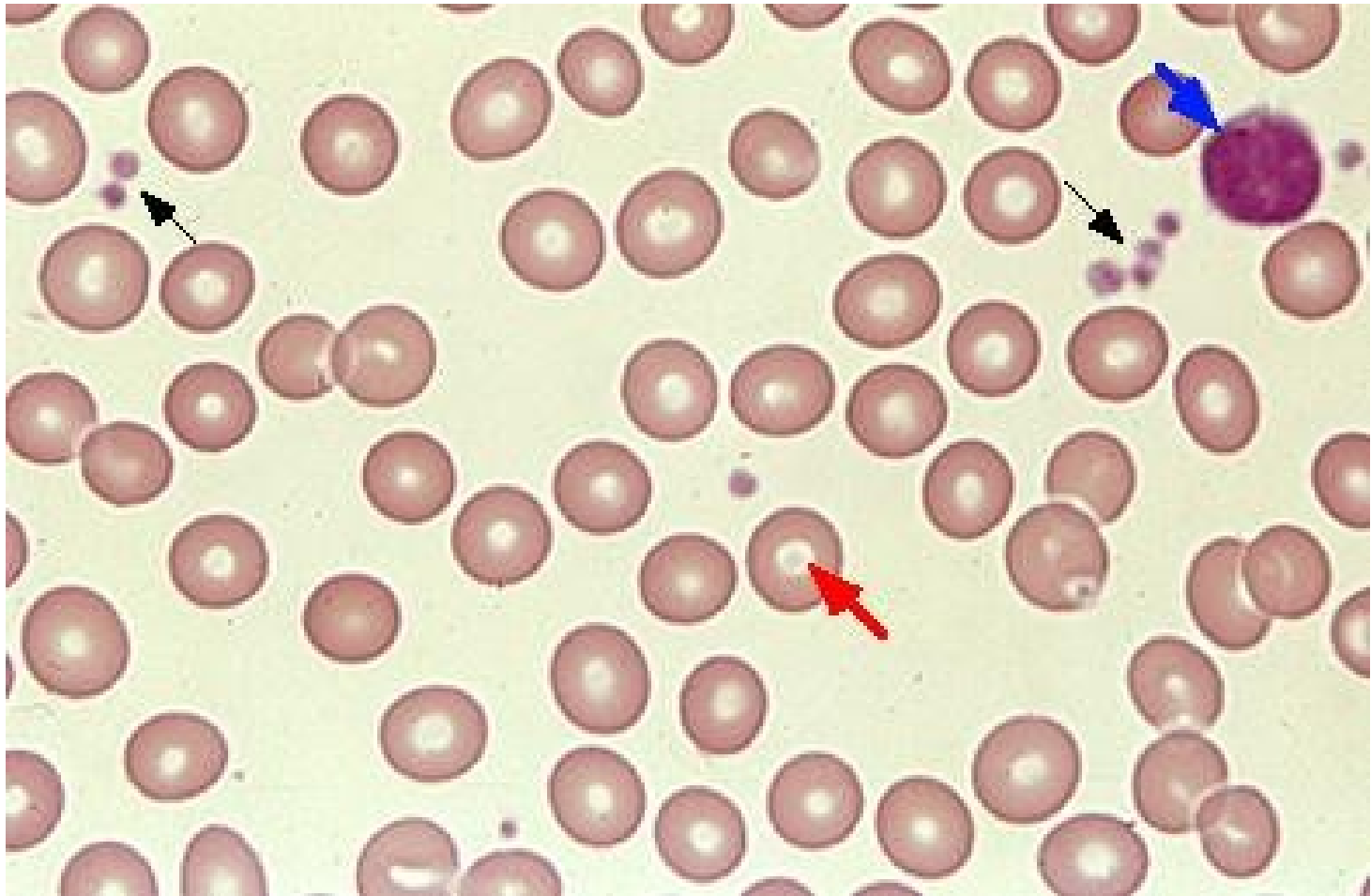
2. Anemias associated with increased RBC production characterized by reticulocytosis

Hemolytic anemias

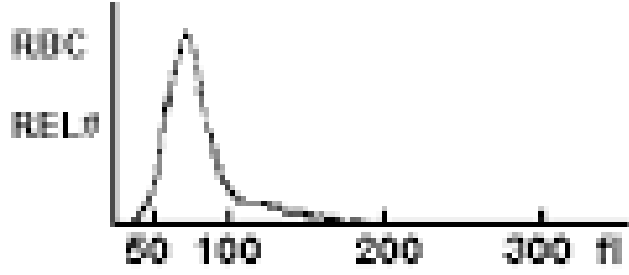
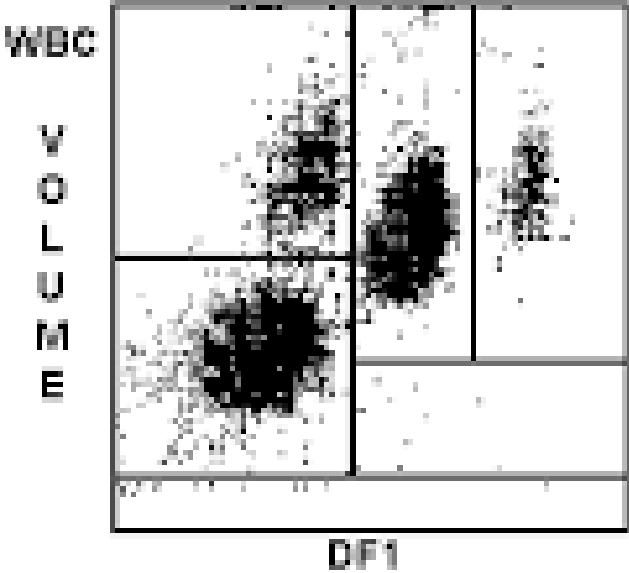
intrinsic –hemoglobinopathy, membrinopathy, enzymopathy

extrinsic – immunologic , hemolytic disease of the newborn, toxins, microangiopathic hemolytic anemias

Normal RBC morphology

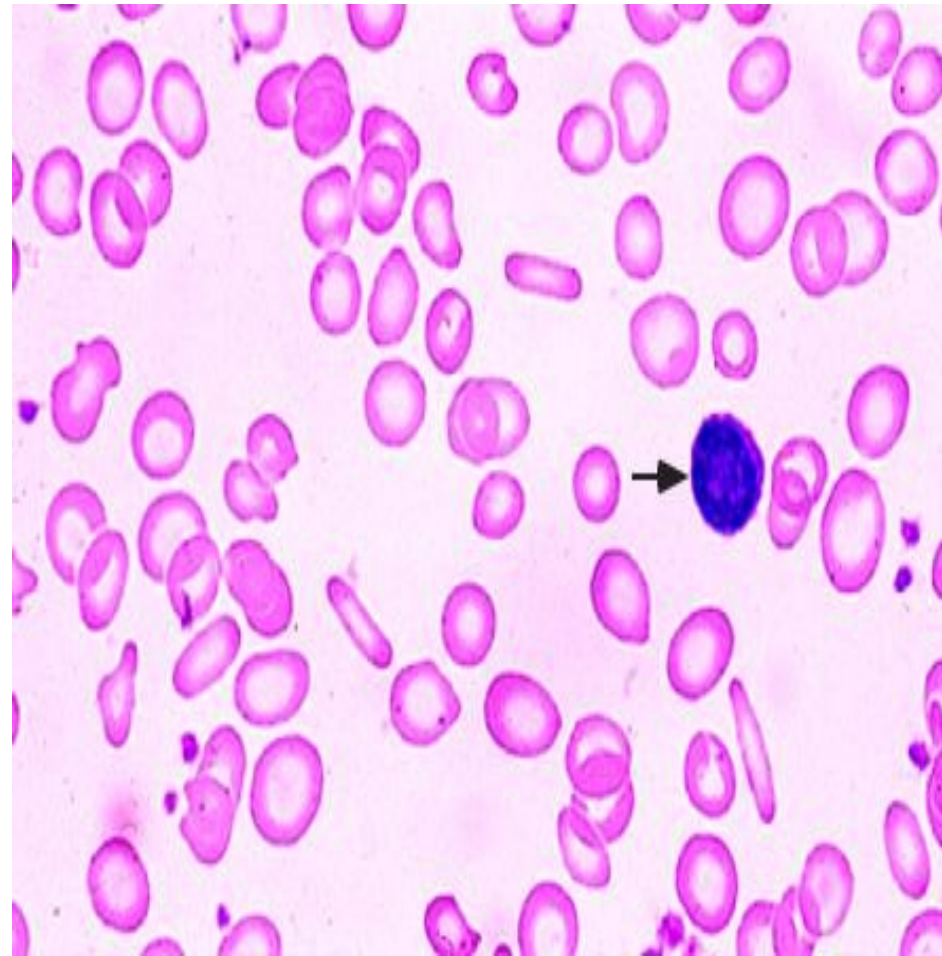


CBC feature



WBC	5.5	
NE	54.7	3.0
LY	34.1	1.9
MO	7.5	0.4
EO	3.0	0.2
BA	0.7	0.0
RBC	4.28	L
HGB	9.7	L
HCT	29.9	L
MCV	69.7	L
MCH	22.6	L
MCHC	32.4	L
RDW	18.4	H
PLT	331	
MPV	8.8	

IDA



Microcytic hypochromic RBC

Tear drop cells

Pencil shaped

Target cells may be seen

DDX

- α - or β -thalassemia
- and other hemoglobinopathies, E and C
- anemia of inflammation
- Lead poisoning

Cause

- Deficient intake
- Growth
- Blood loss
- Impaired absorption
- Inadequate presentation to erythroid precursors
- Abnormal intracellular transport/utilization

Stages of Iron depletion

- **Iron depletion:** may be detected by low serum ferritin
- **Iron-deficient erythropoiesis:** serum iron level drops and the total iron binding capacity increases without a change in the hematocrit. Erythropoiesis begins to be limited by a lack of available iron and serum transferrin receptor levels increase.
- **Iron-deficiency anemia:** is associated with erythrocyte microcytosis, hypochromia, increased RDW and elevated levels of FEP

manifestation

– GI

- anorexia
- pica
- atrophic glossitis
- guaiac positive stool

Cardiac

increased cardiac output
cardiac hypertrophy

Musculoskeletal

impaired exercise performance

– CNS

- fatigue
- Irritability
- Poor development
- trombosis

Labs of IDA

In IDA only RDW
Total Iron Binding
capacity, FEP and
transferrin
receptor are
increased

Key feature is that
Ferritin is low

STUDY	IRON DEFICIENCY ANEMIA	α OR β THALASSEMIA	ANEMIA OF CHRONIC DISEASE
Hemoglobin	Decreased	Decreased	Decreased
MCV	Decreased	Decreased	Normal-decreased
RDW	Increased	Normal	Normal-increased
RBC	Decreased	Normal-increased	Normal-decreased
Serum ferritin	Decreased	Normal	Increased
Total Fe binding capacity	Increased	Normal	Decreased
Transferrin saturation	Decreased	Normal	Decreased
FEP	Increased	Normal	Increased
Transferrin receptor	Increased	Normal	Increased
Reticulocyte hemoglobin concentration	Decreased	Normal	Normal-decreased

FEP, free erythrocyte protoporphyrin; MCV, mean corpuscular volume; RBC, red blood cell; RDW, red cell distribution width.

- Treatment

- Oral iron administration 4-6 mg/kg of elemental iron in 3 divided dose
for 1-3mo in order to replenish the store
- Dietary education of the parents
- Parenteral iron limited role
- Blood transfusion is indicated when the anemia is severe $Hgb < 4g/dl$, presence of failure

Time after administration

Response

- Ⓐ 12-24hrs----- intracellular enzyme replaced
(subjective improvement: good appetite, decrease irritability)
- Ⓐ 36-48hrs-----initial BM response; erythroid hyperplasia
- Ⓐ 48-72hrs-----Reticulocytosis
- Ⓐ 4-30days-----increase in Hb
- Ⓐ 1-3months-----Store replacement

Thalassemia

Inherited

Major and Minor

Major seen after 3-6 months

C/f

Anemia at 6 months

Jaundice

Frontal Bossing

Splenomegally

##Target cells are characteristic

Tx- periodic blood transfusion ,
splenectomy and folic supplementation

Sideroblastic

Cerebellar sx- like lack of coordination,
ataxia

DTR

Incoordination

#CXCS- Increased serum iron and
ferritin with decreased TIBC

Opposite of IDA

Peripheral smear shows papenhemier
bodies in RBC

Ringed sideroblast is hallmark

Tx- Vitamin B6(pyridoxine)

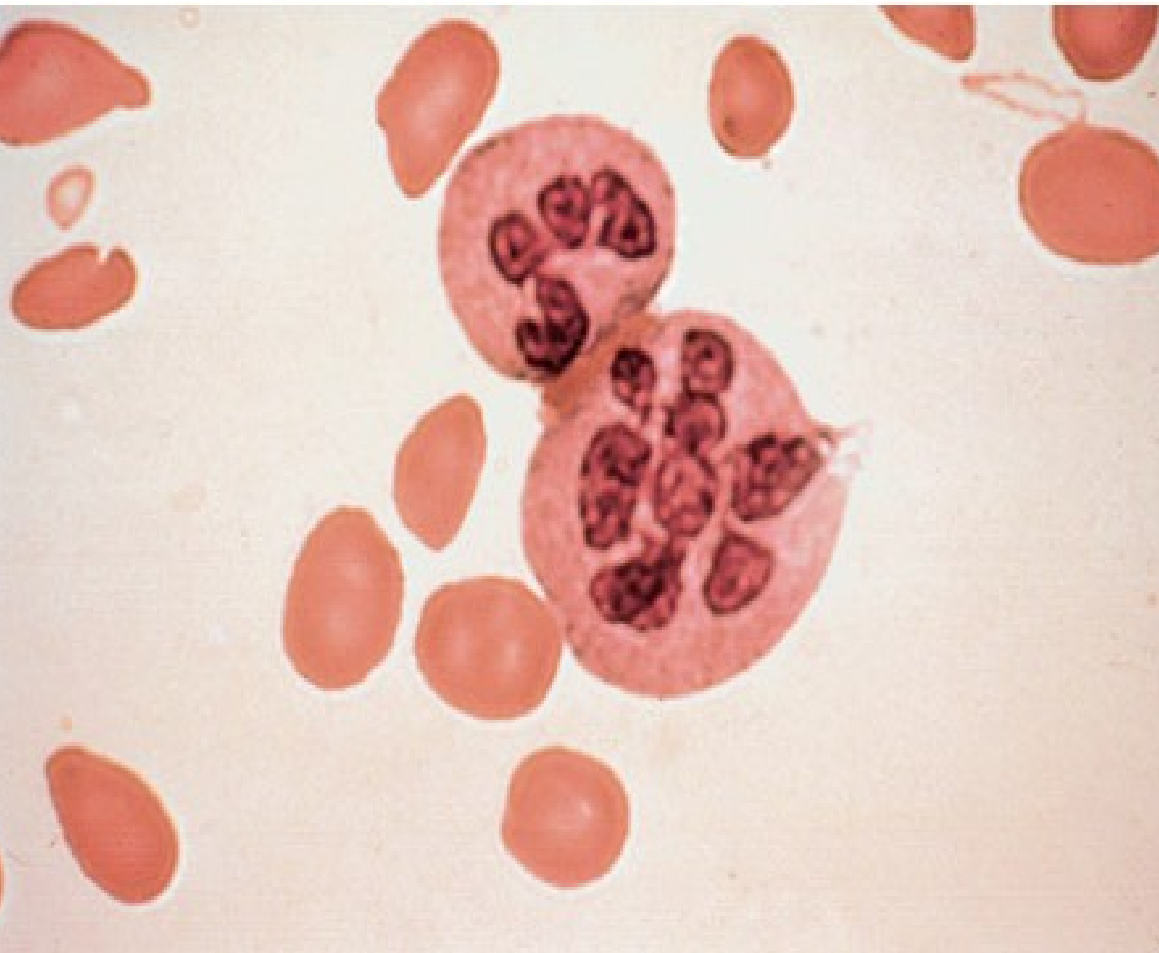
Important

Cause of Anemia	Serum Iron Level	Total Iron-Binding Capacity	Percent Saturation	Soluble Serum Transferrin Receptor Level	Bone Marrow Storage Iron
Iron deficiency	Decreased	Increased	Decreased	High	Decreased
Thalassemias	Increased or normal	Decreased or normal	Increased or normal	Variable, may be high	Increased or normal
Sideroblastic anemias	Increased	Decreased or normal	Increased	Variable, may be high	Increased
Chronic disease	Decreased	Decreased	Decreased	Normal	Increased

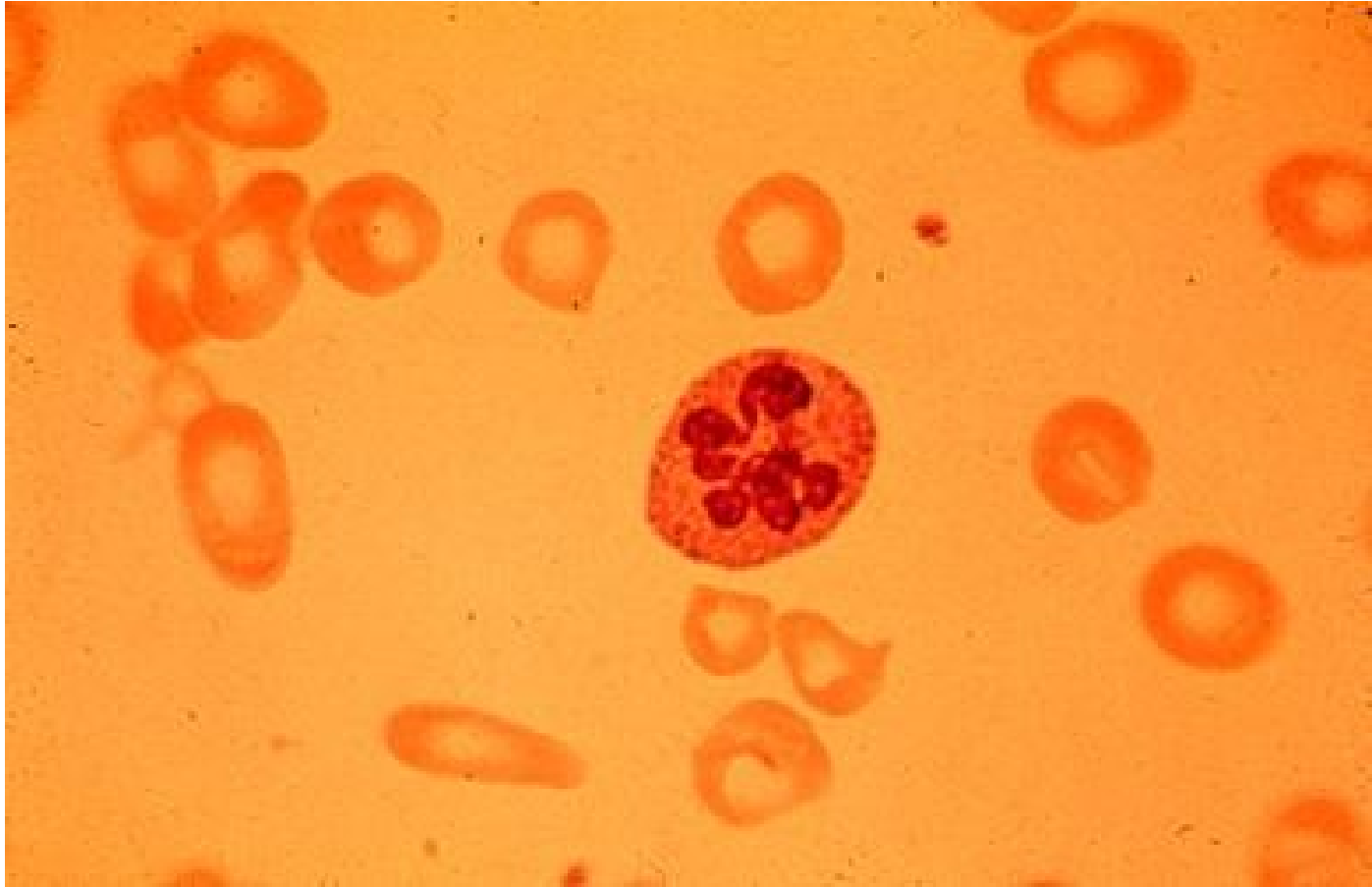
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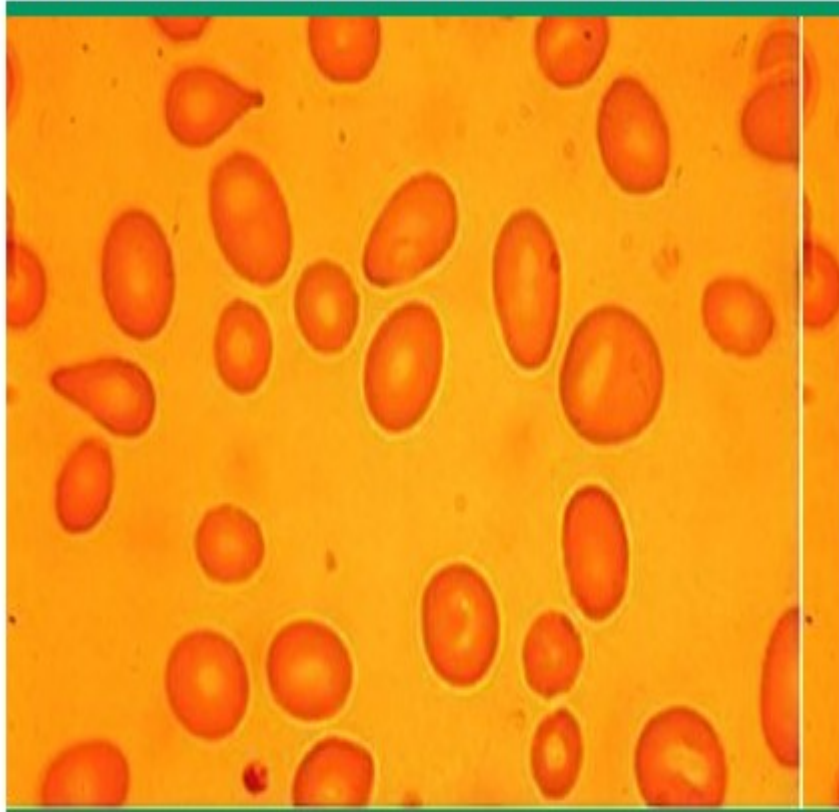
Megaloblastic Anemia



The peripheral blood smear is notable for large, often oval, RBCs, with increased mean corpuscular volume (MCV). Neutrophils are characteristically hypersegmented, with many having >5 lobes

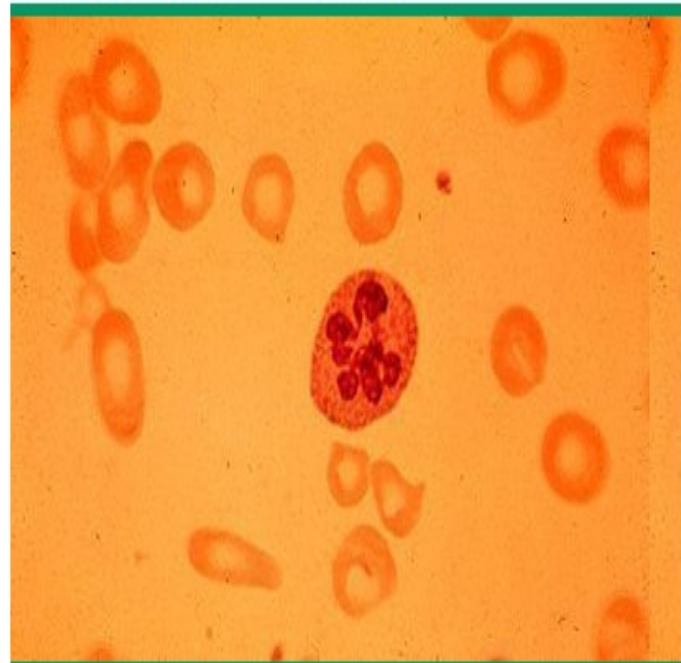


Macroovalocytes in vitamin B12 deficiency



Peripheral smear shows marked macroovalocytosis in a patient with vitamin B12 deficiency.

Megaloblastic blood picture



Peripheral blood smear showing a hypersegmented neutrophil (7 lobes) and macroovalocytes, a pattern that can be seen with cobalamin or folate deficiency.

Folate

- Is abundant in many foods
- Biologically active form is THF
- absorbed throughout the small intestine
- anemia occurs after 2-3 months of folate free diet
- **Etiology**
 - *Inadequate folate intake*
 - *Decreased folate absorption*
 - *Congenital abnormalities in folate metabolism*
 - *Drug induced abnormalities in folate metabolism*
- MX- Folic acid 0.5-1 mg/day for 3-4 wks till definite hematologic response

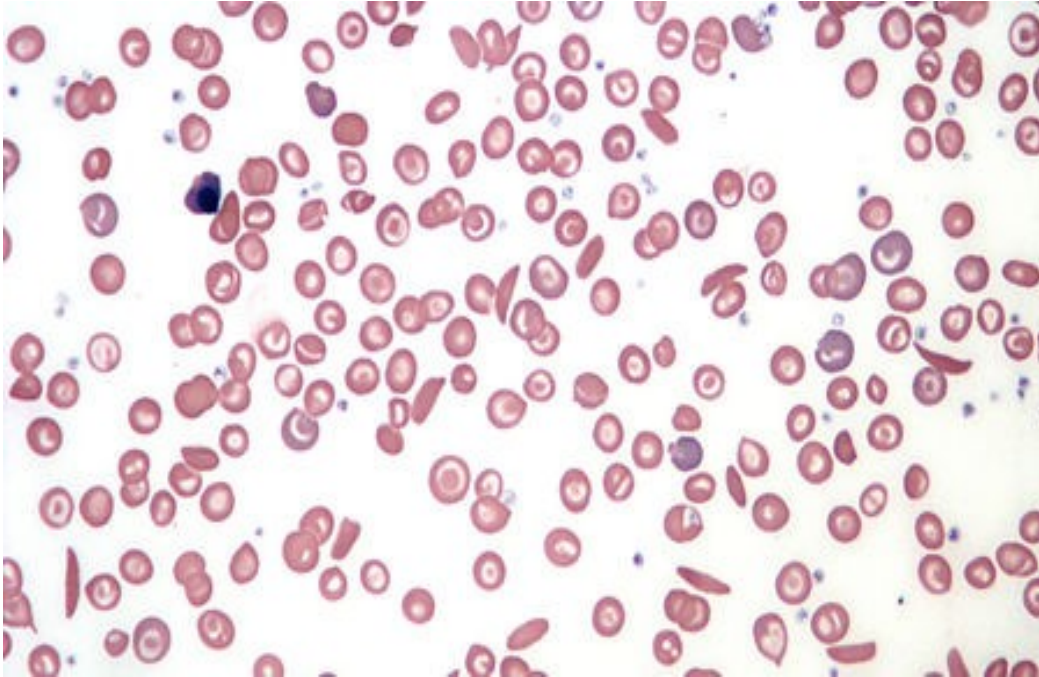
Vit. B12

- Etiology
 - *Inadequate intake*
 - *Lack of intrinsic factor*
 - *Impaired absorption*
- *Absence of transport protein* weakness, fatigue, FTT or irritability
- Other findings include pallor, glossitis, vomiting, diarrhea and icterus
- Neurologic manifestations include parasthesias, sensory deficit, hypotonia, seizure, developmental delay, developmental regression and neuropsychiatric changes

Mx of vit B12

- Prompt hematologic response follows parenteral administration of vit. B12 (**1mg**) usually with reticulocytosis in 2-4 days
- Neurologic manifestation 1 mg should be injected daily for **2 weeks**
- Monthly administration of 1 mg of vit. B12 IM is sufficient as maintenance

Sickle cell anemia



Cxcs of sickle cell

- Low hemoglobin and hct
- High reticulocyte count
- High WBC, PLT and serum bilirubin
- Present as a sudden severe anemia and severe pain, fever – vasoocclusive crisis
- Peripheral Smear- Howel Jolly bodies and sickle shaped RBC
- Tx- BM transplant, hydroxyurea and in the case of hemolytic crisis- IV fluid, oxygen, Blood transfusion

Hemolytic Anemia

- RBC survival is shortened
- the RBC count falls
- erythropoietin is increased
- and the stimulation of marrow activity results in heightened RBC production

Causes of hemolytic anemia

- INTRACORPUSCULAR HEMOLYSIS
 - Membrane Abnormalities
 - Metabolic Abnormalities
 - Hemoglobinopathies
- EXTRACORPUSCULAR HEMOLYSIS
 - Nonimmune
 - Idipathic
 - Secondary
 - Immune
 - Isoimmune
 - Autoimmune
 - Idiopathic
 - Secondary : infection , drugs , hematologic , immunopathic , tumors

- **Laboratory Features**
- **Direcet coombs test is postive**
- **Peripheral blood smear microscopy:**
 - fragments of the red blood cells ("schistocytes") can be present
 - some red blood cells may appear smaller and rounder than usual (spherocytes)
- **Reticulocytes** are present in elevated numbers.
- **Bone marrow smear microscopy: erythroid hyperplasia (megaloblasts)**
- The level of unconjugated bilirubin in the blood is elevated.
- The level of lactate dehydrogenase (LDH) in the blood is elevated

Hemolytic anemia tx

- Treatment
 - Blood transfusion
 - Splenectomy
 - Treat the cause
 - Immunosuppressive

Bleeding Disorder

**Bleeding
disorders**

**Vascular
abnormalities**

Platelet disorders**

**Clotting factor
abnormalities**

DIC

ITP

- classic presentation of ITP is that of a previously healthy 1–4 yr old child who has sudden onset of generalized petechiae and purpura.
- The parents often state that the child was fine yesterday and now is covered with bruises and purple dots. Often there is bleeding from the gums and mucous membranes, particularly with profound thrombocytopenia (platelet count $<10 \times 10^9/L$). There is a history of a preceding viral infection 1–4 wk before the onset of thrombocytopenia.
- Findings on physical examination are normal, other than the finding of petechiae and purpura. Splenomegaly is rare, as is lymphadenopathy or pallor.

CXCS of severity

- No symptoms
- 2. Mild symptoms: bruising and petechiae, occasional minor epistaxis, very little interference with daily living
- 3. Moderate: more severe skin and mucosal lesions, more troublesome epistaxis and menorrhagia
- 4. Severe: bleeding episodes—menorrhagia, epistaxis, melena—requiring transfusion or hospitalization, symptoms interfering seriously with the quality of life

Additional ix

- ANA
- Prolonged BT but normal PT and PTT
- HIV
- CBC with differential
- BM examination
- Coombs
- **Peripheral smear shows thrombocytopenia and giant PLTs**
- **BM- megakaryocytes**

Management of ITP

- If mild to moderate sx- no therapy other than education and counseling
- IV IG -to induce a response by down regulating Fc-mediated phagocytosis of antibody-coated platelets
- IV anti D for RH positive
- Prednisone for acute and chronic ITP in adults
- Chronic ITP- (>6months of thrombocytopenia)
 - Evaluate for other autoimmune conditions
 - Do splenectomy after giving pneumococcal **vaccine**
 - **Feared complication- intracranial bleed**

HSP

- Male patient that presents with a **The hallmark of the disease is the rash, beginning as pinkish maculopapules that initially blanch on pressure and progress to petechiae or purpura, which are characterized clinically as palpable purpura that evolve from red to purple to rusty brown before they eventually fade .**
- **Arthritis, present in more than $\frac{2}{3}$ of children with HSP, is usually localized to the knees and ankles and appears to be concomitant with edema.**
- **Colicky abdominal pain**
- **Hematuria, proteinuria**
- 50% also have a preceeding URTI in the past 3 weeks



Rash

Edema

Arthritis with abdominal pain and renal findings

IX-

Esr- increased

Thrombocytosis, leukocytosis

Anemia

ANA

APL

Urine analysis with WBC, RBC casts, albumin

Renal biopsy showing mesangial deposits

Complications and MX

- Renal involvement - Nephrotic syndrome
- Bowel perforation
- Scrotal edema, testicular torsion
- MX
- Symptomatic treatment

DIC

- Acute DIC- bleeding from venipuncture or surgical sites
- Petechiae and echymoses, necrosis
- Minimal to profound shock, renal failure, dyspnea, convulsion and coma
- Low- PT,PTT, Platelet, fibrinogen
- Elevated plasma fibrin
- Fragmented burr and helmet shaped RBC-schistocytes

Hemophilia A and B

- Factor 8 deficiency - Hem.A
- Factor 9 deficiency- Hem.B
- Males more commonly affected
- Both factors don't cross the placentas so symptoms may be present from birth and may occur in the fetus

- easy bruising, intramuscular hematomas, and hemarthroses
- Bleeding from minor traumatic lacerations of the mouth (a torn frenulum) may persist for h
- The hallmark of hemophilia is hemarthrosis eg. Ankle joint hemorrhage

Hemophilia labs

- PTT-decreased in factor 8
- CT and PTT in factor 9
- Platelet, bleeding time and prothrombin time are all normal
- Dx- decreased factor level with normal von willebrand factor antigen

Tx of hemophilia

- Early, appropriate therapy is the hallmark of excellent hemophilia care.

Prevention of trauma

Avoid aspirine

immunization

- With mild factor VIII hemophilia, the patient's endogenously produced factor VIII can be released by the administration of desmopressin acetate

Hemorrhagic disease of the newborn

- 1.Vitamin K
 - Spontaneous hemorrhage in a new born
 - Bleeding time, fibrinogen, factors V and VIII, platelets, capillary fragility, and clot retraction are normal for maturity

TABLE 103-4 -- Hemorrhagic Disease of the Newborn

	EARLY ONSET	CLASSIC DISEASE	LATE ONSET
Age	0–24 hr	2–7 days	1–6 mo
Site of hemorrhage	Cephalohematoma	Gastrointestinal	Intracranial
	Subgaleal	Ear-nose-throat–mucosal	Gastrointestinal
	Intracranial	Intracranial	Cutaneous
	Gastrointestinal	Circumcision	Ear-nose-throat–mucosal
	Umbilicus	Cutaneous	Injection sites
	Intra-abdominal	Gastrointestinal	Thoracic
		Injection sites	
Etiology/risks	Maternal drugs (phenobarbital, phenytoin, warfarin, rifampin, isoniazid) that interfere with vitamin K	Vitamin K deficiency	Cholestasis—malabsorption of vitamin K (biliary atresia, cystic fibrosis, hepatitis)
	Inherited coagulopathy	Breast-feeding	Abetalipoprotein deficiency
			Idiopathic in Asian breast-fed infants
			Warfarin ingestion
Prevention	Possible vitamin K at birth or to mother (20 mg) before birth	Prevented by parenteral vitamin K at birth. Oral vitamin K regimens require repeated dosing over time	Prevented by parenteral and high-dose oral vitamin K during periods of malabsorption or cholestasis
	Avoid high-risk medications		
Incidence	Very rare	≈2% if not given vitamin K	Dependent on primary disease

PT,PTT & CT are prolonged

Level of factor II,VII,IX AND X are prolonged.

BT, factor I,V and platelets are normal.

Diagnosis is confirmed by demonstration of
PIVKA and improvement with vitamin k
treatment.

MX-

IM 1mg vitamin K- prophylactic

Treatment – slow IV infusion of vit.K

Newborn DIC

Affected infants are often premature; the clinical course is frequently characterized by asphyxia, hypoxia, acidosis, shock, hemangiomas, or infection.

Results secondary to an insult eg. maternal toxemia, GBS infection, Abruptio placenta, Severe RDS, Necrotizing enterocolitis, congenital viral disease(CMV and HSV), Fetal demise and Erythroblastosis fetalis

MX of newborn DIC and investigations

- Ix- low platelet, increased factor 1 and FDP
- MX- treat the cause

Other HDN d/o

- Swallowed blood syndrome
 - Blood or bloody stool usu passed on the 2nd or 3rd day
 - Differentiation from gastrointestinal hemorrhage is based on the fact that the infant's blood contains mostly fetal hemoglobin, which is alkali-resistant,
 - whereas swallowed blood from a maternal source contains adult hemoglobin, which is promptly changed to alkaline hematin after the addition of alkali. Apt test is used for this differentiation:

Table 6. Common Bleeding Disorders and Basic Laboratory Findings

DISORDER	PLATELET COUNT	PT	APTT
Thrombocytopenia	Low	Normal	Normal
Platelet dysfunction	Normal	Normal	Normal
Hemophilia	Normal	Normal	Prolonged
FVII deficiency	Normal	Prolonged	Normal
vWD	Usually normal	Normal	Normal to prolonged
Vitamin K deficiency	Normal	Prolonged	Prolonged
Liver disease	Low to normal	Prolonged	Prolonged
DIC	Low	Prolonged	Prolonged
Vascular anomaly	Normal	Normal	Normal

Key: DIC – disseminated intravascular coagulation; PT – prothrombin time; aPTT – activated partial thromboplastin time